

UNIFORMITY, PRIME SUPPORT, AND REACHABLE LATTICES IN APPROACHES TO THE *abc* CONJECTURE

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ABSTRACT. We examine precise uniformity requirements in several approaches to the *abc* conjecture. An elementary encoding bounds the number of integers $n \leq N$ with all prime divisors at most Y and at most w distinct prime divisors by $(1 + Y \lfloor \log_2 N \rfloor)^w$. Combined with an unconditional short-interval theorem of Younis, this shows that proposed smooth-number moment estimates in a recent claimed disproof are false in their stated parameter range. We also give the exact coupled prime-exponent formulation of the *abc* height defect and an unbounded dyadic obstruction to estimating its endpoints separately. For local tensor constructions, we prove a finite determinant certificate for the Haar measure of the lattice generated by reachable vectors, and distinguish independent actions from simultaneous ones. Effective integral-point bounds constrain a specified Pell–Chebyshev residual family, while an explicit lower bound for the denominator of the Frey j -invariant limits the applicability of recent small-denominator results. Elementary arithmetic, counting, and linear-algebra components are checked in Lean; external analytic inputs and remaining interfaces are identified separately. Neither a proof nor a disproof of *abc* is claimed.

1. THE TARGET AND THE ROLE OF THE RESULTS

Throughout, an *abc triple* means positive integers a, b, c such that $a + b = c$ and a, b, c are pairwise coprime. Write

$$r = \text{rad}(abc), \quad R = \log r, \quad h = \log c.$$

Here $\text{rad}(n)$ is the product of the distinct prime divisors of n . Pairwise coprimality is equivalent to $\gcd(a, b, c) = 1$ under the additive relation. The target is the standard assertion

$$(1) \quad \forall \varepsilon > 0 \quad \exists C_\varepsilon \in \mathbb{R} \quad \forall (a, b, c) : \quad h \leq (1 + \varepsilon)R + C_\varepsilon.$$

Exponentiation gives the usual multiplicative constant. The repository’s `ABCConjecture` uses $\log \max(a, b, c)$, which equals h here. Its definition is not changed in this work.

The quantifier order in (1) is essential. Its negation requires one fixed positive ε and violations for every additive constant, hence at unbounded height. A finite set of high-quality triples cannot supply that negation: it is absorbed into C_ε . Likewise, a sparse exceptional-set bound does not imply that the exceptional set is finite.

Our results concern four specific interfaces. Section 2 uses finite prime support to test a proposed analytic construction. Section 3 records the cancellation that an endpoint approach must preserve. Section 4 supplies a local generator certificate relevant to possible-image constructions. Sections 5 and 6 examine effective arithmetic geometry and recent literature. These components are useful tests and partial theorems; none constructs the missing uniform bound in (1).

The new statements below were proved mathematically before their corresponding Lean implementations. Independent cross-reviews checked the arguments, source hypotheses, and formal scope. Classical published theorems may be used in the paper, but are not silently represented as proved Lean declarations. The formal scope is specified in Section 7. The literature cutoff is August 30, 2026.

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2. PRIME SUPPORT IN SHORT INTERVALS

Let $\omega(n)$ be the number of distinct prime divisors of n , and let $P^+(n)$ be its greatest prime divisor, with $P^+(1) = 1$. For integers $N, Y \geq 1$ and $w \geq 0$, define

$$\mathcal{A}(N, Y, w) = \{1 \leq n \leq N : P^+(n) \leq Y, \omega(n) \leq w\}.$$

Theorem 2.1 (Finite prime-power encoding). *With $L = \lfloor \log_2 N \rfloor$, one has*

$$(2) \quad |\mathcal{A}(N, Y, w)| \leq (1 + YL)^w.$$

Proof. Use the alphabet

$$\mathcal{Q} = \{1\} \cup \{q^e : 1 \leq q \leq Y, 1 \leq e \leq L\}.$$

It has at most $1 + YL$ members; the bases need not be prime. For $n \in \mathcal{A}(N, Y, w)$, write its actual prime factorization as $n = \prod_{j=1}^t p_j^{e_j}$, with $p_1 < \dots < p_t$ and $t \leq w$. Since $2^{e_j} \leq p_j^{e_j} \leq n \leq N$, each $e_j \leq L$. Thus every factor belongs to $\mathcal{Q}$. Append $w - t$ copies of 1 to obtain a word of length w . Its product recovers n , so different integers have different encoded words. Counting all words proves (2). The empty factorization covers $n = 1$ and $w = 0$; $N = 1$ is also included. $\square$

Corollary 2.2 (A finite first-moment bound). *Let S be a finite set of distinct positive Y -smooth integers at most N , and put $B = (1 + Y \lfloor \log_2 N \rfloor)^w$. Then*

$$(3) \quad |\{n \in S : \omega(n) \leq w\}| \leq B,$$

$$(4) \quad \sum_{n \in S} \omega(n) \geq (w + 1) \max\{|S| - B, 0\}.$$

Proof. The first set is contained in $\mathcal{A}(N, Y, w)$. Its complement in S has at least $\max\{|S| - B, 0\}$ members, each with at least $w + 1$ distinct prime factors. Summation proves the second assertion. $\square$

Suppose now that $y = y(x) \geq 2$ satisfies

$$(5) \quad \log \log x = o(\log y), \quad \log y = o(\log x), \quad u = \frac{\log x}{\log y}.$$

For every fixed $\kappa > 0$, Theorem 2.1, with $N = \lfloor 2x \rfloor$, $Y = \lfloor y \rfloor$ and $w = \lfloor \kappa u \rfloor$, gives

$$(6) \quad \#\{n \leq 2x : P^+(n) \leq y, \omega(n) \leq \kappa u\} \leq x^{\kappa + o(1)}.$$

Indeed,

$$w \log(1 + YL) \leq \kappa \frac{\log x}{\log y} (\log y + O(\log \log x)) = \kappa \log x + o(\log x).$$

More generally the upper bound is $x^{o(1)}$ whenever $w(\log y + \log \log x) = o(\log x)$.

2.1. The external analytic input. Write $\Psi(x, y) = \#\{1 \leq n \leq x : P^+(n) \leq y\}$. Younis's unconditional Theorem 1.1 [16], p. 2, states that for each fixed $17/30 < \theta \leq 1$ there is $C_Y(\theta) > 0$ such that

$$(7) \quad \frac{\Psi(x + h, y) - \Psi(x, y)}{h} = \frac{\Psi(x, y)}{x} \left(1 + O_\theta \left(\frac{\log(u + 1)}{\log y} \right) \right)$$

uniformly for

$$x^\theta \leq h \leq x, \quad \exp\left(C_Y(\theta)(\log x)^{2/3}(\log \log x)^{4/3}\right) \leq y \leq 2x.$$

Fix

$$(8) \quad \frac{17}{30} < \theta < 1, \quad C \geq C_Y(\theta), \quad h = x^\theta, \quad y = \exp\left(C(\log x)^{2/3}(\log \log x)^{4/3}\right).$$

The lower bound on C is part of the hypotheses, not a disposable constant. Put

$$S_x = \{n \in (x, x + h] : P^+(n) \leq y\}, \quad M_x = |S_x|.$$

The classical long-interval estimate recalled in [5, equations (1.8)–(1.9), p. 415] yields $\Psi(x, y) = x\rho(u)(1 + o(1))$ here. Its range $u \leq (\log y)^{3/5 - \varepsilon}$ is satisfied, for example with $\varepsilon = 1/20$: the power

of $\log x$ on the right is $11/30$, larger than the power $1/3$ in u . Also $-\log \rho(u) = u \log u(1+o(1)) = o(\log x)$. Consequently (7) gives

$$(9) \quad M_x = h\rho(u)(1+o(1)) = x^{\theta-o(1)}.$$

Only this mass estimate is needed below; no short-interval moment formula is assumed.

Theorem 2.3 (A lower tail and a first-moment obstruction). *For (8) and every fixed $0 < \kappa < \theta$,*

$$(10) \quad \frac{\#\{n \in S_x : \omega(n) \leq \kappa u\}}{M_x} \leq x^{\kappa-\theta+o(1)} \rightarrow 0.$$

In particular,

$$(11) \quad \liminf_{x \rightarrow \infty} \frac{1}{uM_x} \sum_{n \in S_x} \omega(n) \geq \theta.$$

For every fixed $A > 0$, the proportion of $n \in S_x$ with $\omega(n) \leq (\log \log x)^A$ tends to zero.

Proof. Every member of S_x is at most $2x$. Divide (6) by (9) to obtain (10). If its numerator is B_x , then

$$\sum_{n \in S_x} \omega(n) \geq \kappa u(M_x - B_x).$$

The lower limit after division by uM_x is at least each fixed $\kappa < \theta$, and hence at least θ . This argument does not require a variable κ or a uniform error term in that variable. Finally,

$$(\log \log x)^A (\log y + \log \log x) = o(\log x).$$

The last observation after (6) bounds the corresponding population by $x^{o(1)}$. Divide again by (9). $\square$

Corollary 2.4 (A false intermediate assertion in a claimed disproof). *Lemma 4.2 of Carella [4], p. 8, equation (4.5), is false in its stated range. The same holds for the second-moment assertion in Lemma 4.4, p. 9, and the relative exceptional estimate (4.24), p. 11.*

Proof. Choose $\theta = 3/5$ and a fixed $C \geq C_Y(3/5)$. Then $h > x^{7/12}$, $y < h < x$, and $y = o(h)$, exactly as required in the first lemma. Its proposed formula would give

$$\frac{1}{M_x} \sum_{n \in S_x} \omega(n) = O(\log \log y).$$

However,

$$\frac{u}{\log \log y} \asymp \frac{(\log x)^{1/3}}{(\log \log x)^{7/3}} \rightarrow \infty,$$

contradicting (11). Changing the left endpoint of the interval from open to closed adds at most one integer; its contribution is at most $\log_2(2x)$ and is negligible relative to M_x . The proposed second moment is $O(M_x(\log \log y)^2)$, whereas (10) gives a lower bound $\kappa^2 u^2 M_x(1 - o(1))$ for any fixed $0 < \kappa < 3/5$. Finally the threshold in (4.24) is bounded by a fixed power of $\log \log x$. Theorem 2.3 shows that the proportion above it tends to one, although the claimed relative upper bound tends to zero. $\square$

Remark 2.5. This refutes those intermediate assertions, not *abc* and not the possibility of an exceptionally sparse counterexample family. In particular a set of size $x^{o(1)}$ can still be infinite. It is also important to distinguish $o(h)$ from $o(M_x)$: the whole smooth set already has size $o(h)$.

2.2. Two further transfer tests. For a fixed integer $k \geq 4$, the intervals

$$(p^k, p^k + (p^k)^{3/5}], \quad X < p^k \leq 2X,$$

are pairwise disjoint for large X . Indeed, successive distinct integer bases give gaps at least $X^{1-1/k}$, larger than $(2X)^{3/5}$. The finite encoding, using the largest thresholds in this range, shows that at most $X^{o(1)}$ centres can contain a smooth candidate with $\omega(n) \leq 2 \log \log(p^k)$ at the scale (8). The prime number theorem gives $X^{1/k+o(1)}$ centres, so their proportion is at most $X^{-1/k+o(1)}$. This rules out a positive-proportion selection of this particular kind, while leaving a sparse selection unresolved.

Recent work on exceptional sets [2, 3] and on the existence of smooth neighbours [6] addresses different quantifiers. A useful general check is the following. If a counting theorem gives $N(T) \ll T^{\eta+o(1)}$, then a seed of height H contradicts it only when it produces more than $H^{\beta\eta+o(1)}$ *distinct* qualifying outputs of height at most $O(H^\beta)$. Repeated encodings of one output do not count. None of the existence results just cited supplies the required uniform low-radical neighbours or such an amplification theorem.

3. THE COUPLED SIGNED PRIME-EXPONENT DEFECT

For $n \geq 1$ put

$$E_1(n) = \sum_{\substack{p|n \\ v_p(n)=1}} \log p, \quad E_3(n) = \sum_{p|n} (v_p(n) - 2)_+ \log p.$$

Write $m = \min(a, b)$ and $M = \max(a, b)$ for an abc triple.

Proposition 3.1 (Exact signed identity). *For every positive integer n ,*

$$(12) \quad \delta_2(n) := \log n - 2 \log \text{rad}(n) = E_3(n) - E_1(n).$$

The coupled defect satisfies

$$(13) \quad \begin{aligned} \Delta &:= \log(Mc) - 2R \\ &= E_3(M) + E_3(c) - E_1(M) - E_1(c) - 2 \log \text{rad}(m), \end{aligned}$$

$$(14) \quad 2h - 2R - \log 2 \leq \Delta \leq 2h - 2R.$$

Consequently abc is equivalent to

$$(15) \quad \forall \varepsilon > 0 \quad \exists K_\varepsilon \quad \forall (a, b, c) : \quad \begin{aligned} E_3(M) + E_3(c) &\leq E_1(M) + E_1(c) \\ &\quad + 2 \log \text{rad}(m) + 2\varepsilon R + K_\varepsilon. \end{aligned}$$

Proof. For every positive exponent e , $e - 2 = (e - 2)_+ - \mathbf{1}_{e=1}$. Multiply by $\log p$ and sum over the actual prime factorization to obtain (12). Pairwise coprimality gives $r = \text{rad}(m) \text{rad}(M) \text{rad}(c)$, which proves (13). Since $c/2 \leq M \leq c$, taking logarithms of $c^2/2 \leq Mc \leq c^2$ gives (14). An abc constant C_ε supplies $K_\varepsilon = 2C_\varepsilon$. Conversely (15) and the lower corridor give $C_\varepsilon = (K_\varepsilon + \log 2)/2$. $\square$

The equivalence does not prove (15); it identifies the entire missing estimate without losing the negative exponent-one terms. Replacing the coupled inequality by separate endpoint bounds fails.

Proposition 3.2 (An unbounded separate-endpoint obstruction). *For every $0 \leq \rho < 1$ and $K \in \mathbb{R}$, there is an abc triple with $m = 1$ and*

$$\delta_2(M) > \log \text{rad}(m) + \rho R + K.$$

Proof. Take $(a, b, c) = (1, 2^N, 2^N + 1)$ with $N \geq 1$. Then $m = 1$, $M = 2^N$, and $\delta_2(M) = (N - 2) \log 2$. Without making any assumption about the odd neighbour,

$$r \leq 2(2^N + 1) \leq 2^{N+2}, \quad R \leq (N + 2) \log 2.$$

Therefore

$$\delta_2(M) - \rho R \geq ((1 - \rho)N - 2 - 2\rho) \log 2 \longrightarrow \infty.$$

This proves the claim. The bound uses only $\text{rad}(n) \leq n$ and $\text{rad}(2^N) = 2$. $\square$

This argument neither proves nor assumes that all odd neighbours have large radical. It only rejects a false stronger estimate that discards their possible contribution. The coupled route remains open.

4. FINITE CERTIFICATES FOR REACHABLE LOCAL LATTICES

Let K be a nonarchimedean local field, $\mathcal{O}$ its valuation ring, π a uniformizer, and $k = \mathcal{O}/\pi\mathcal{O}$ its residue field, with $|k| = q > 1$ and $v(\pi) = 1$. Fix $N \geq 1$, write $L = \mathcal{O}^N$, and normalize additive Haar measure on K^N by $\mu(L) = 1$. For any set $\mathcal{R} \subset L$, let

$$H(\mathcal{R}) = \overline{\text{span}_{\mathcal{O}}(\mathcal{R})}.$$

If nonsingular N -tuples exist in $\mathcal{R}$, define

$$d(\mathcal{R}) = \min_{\substack{r_1, \dots, r_N \in \mathcal{R} \\ \det[r_1 \ \dots \ r_N] \neq 0}} v(\det[r_1 \ \dots \ r_N]);$$

otherwise set $d(\mathcal{R}) = \infty$.

Theorem 4.1 (A finite witness for the exact hull measure). *With $q^{-\infty} = 0$, one has*

$$\mu(H(\mathcal{R})) = q^{-d(\mathcal{R})}.$$

If $d(\mathcal{R}) < \infty$, some matrix A whose columns belong to $\mathcal{R}$ satisfies

$$v(\det A) = d(\mathcal{R}), \quad H(\mathcal{R}) = A\mathcal{O}^N.$$

Proof. In the full-rank case the nonzero determinant valuations are nonnegative integers, so choose A attaining their minimum d . For $r \in \mathcal{R}$, Cramer's rule gives

$$(A^{-1}r)_i = \frac{\det A_i(r)}{\det A},$$

where $A_i(r)$ replaces column i by r . Its columns still belong to $\mathcal{R}$. Every nonzero numerator has valuation at least d ; a zero numerator causes no difficulty. Thus $A^{-1}r \in \mathcal{O}^N$ and $\mathcal{R} \subset A\mathcal{O}^N$. Since the columns of A lie in $\mathcal{R}$, their integral spans are equal. The set $A\mathcal{O}^N$ is compact, hence closed.

Integral row and column operations give $UAV = \text{diag}(\pi^{a_1}, \dots, \pi^{a_N})$, with $U, V \in \text{GL}_N(\mathcal{O})$, $a_i \geq 0$, and $\sum a_i = d$. One may obtain this diagonalization by repeatedly moving an entry of least valuation to the upper left and clearing its row and column by integral operations. The quotient $\mathcal{O}^N/A\mathcal{O}^N$ has q^d elements. Its equal-measure cosets partition L , proving $\mu(A\mathcal{O}^N) = q^{-d}$.

In the rank-deficient case a nonzero K -linear functional annihilates $\mathcal{R}$. Scale its coefficients to lie in $\mathcal{O}$ with at least one unit coefficient. It then defines a surjection $\ell : L \rightarrow \mathcal{O}$ whose closed kernel contains $H(\mathcal{R})$. For every $t \geq 1$, $\ell^{-1}(\pi^t\mathcal{O})$ has index q^t in L . Hence $0 \leq \mu(H(\mathcal{R})) \leq q^{-t}$ for all t , proving measure zero. $\square$

Corollary 4.2. *The equality $H(\mathcal{R}) = L$ holds if and only if the reductions of $\mathcal{R}$ span k^N , equivalently if some reachable column matrix has unit determinant. More generally, any measurable $\mathcal{O}$ -submodule containing all columns of a reachable matrix A with $v(\det A) = d$ has measure at least q^{-d} .*

Proof. A determinant is a unit exactly when its reduction is nonzero; a spanning set of k^N contains a basis. Apply Theorem 4.1. For the last claim, the target contains $A\mathcal{O}^N$ and measure is monotone. The target need not be contained in L . $\square$

4.1. Independent and simultaneous actions. Let $L_j = \mathcal{O}^{d_j}$, with $d_j \geq 1$, and let $T = \bigotimes_{j=1}^s L_j$, where $s \geq 1$ and $N = \prod_j d_j$. For nonzero $x_j \in L_j$, write $x_j = \pi^{m_j} u_j$ with u_j primitive.

Proposition 4.3. *Under the independent action of $\prod_j \mathrm{GL}_{d_j}(\mathcal{O})$, the orbit of $\bigotimes_j x_j$ spans $\pi^{\sum_j m_j} T$. The measure of this spanned lattice, normalized by $\mu(T) = 1$, is $q^{-N \sum_j m_j}$.*

Proof. Integral row operations send a primitive vector to any chosen standard basis vector. Choosing these operations independently in each factor gives every standard tensor basis vector multiplied by $\pi^{\sum_j m_j}$. Conversely all orbit vectors lie in this scaled lattice. Its index is $q^{N \sum_j m_j}$, giving the measure. $\square$

Proposition 4.4 (A strict simultaneous-action counterexample). *Over any commutative ring A , identify $A^2 \otimes_A A^2$ with $\mathrm{Mat}_2(A)$. The A -span of $\{v \otimes v : v \in A^2\}$ is exactly the submodule of symmetric matrices. If $A \neq 0$, it omits E_{12} . For $A = \mathcal{O}$ it has Haar measure zero in $\mathrm{Mat}_2(K)$.*

Proof. Every generator has equal off-diagonal entries, a condition preserved by linear combinations. Conversely $e_1 \otimes e_1, e_2 \otimes e_2$, and

$$(e_1 + e_2) \otimes (e_1 + e_2) - e_1 \otimes e_1 - e_2 \otimes e_2 = E_{12} + E_{21}$$

span all symmetric matrices. No division by two occurs. The functional $M \mapsto M_{12} - M_{21}$ vanishes on the span and takes value one on E_{12} . In the local-field case, the congruence $M_{12} = M_{21} \pmod{\pi^t}$ occupies a proportion q^{-t} of the integral matrix lattice for every t . Its exact equality locus therefore has measure zero. $\square$

The generating set in Proposition 4.4 is stable under every simultaneous action $g \otimes g$. Thus even the full simultaneous integral linear group does not provide Proposition 4.3. This rejects a specific abstract substitute for independent generation; it does not identify the actual IUT outputs with this diagonal example.

4.2. What a source-derived application must establish. The July 2026 LANA report [10] isolates an identification between q-pilot and possible-output volume data in the passage to IUT III, Corollary 3.12 [11]. Joshi's July 30 response [9] refers to distinct arithmeticoids and his Part IV comparison [8, Theorem 6.10.1]. These are source claims to reconstruct, not assumptions used in our local theorems.

Part III [7, Proposition 9.7.5.1] allows specified topological isomorphisms in its collation. It would therefore be inaccurate to say that it permits no automorphism enlargement. However, the simultaneous actions and the product-to-tensor hull operation in Theorem-Definition 9.8.1.1 do not, by themselves, state unrestricted independent integral linear actions. One must prove which choices are permitted and that the actual target contains their integral span.

For a source-faithful finite packet of local sets $\mathcal{R}_v \subset \mathcal{O}_v^{N_v}$ and positive weights λ_v , Theorem 4.1 gives, when all defects are finite,

$$V_H = - \sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v.$$

Suppose genuine local target regions Θ_v contain the respective hulls and have finite positive measure. Define, on these same carriers, $V_\Theta = \sum_v (\lambda_v / N_v) \log \mu_v(\Theta_v)$. Let $V_q \in \mathbb{R}$ be the fixed native q-pilot log-volume under the same source normalization. Then

$$\sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v \leq -V_q \implies V_q \leq V_H \leq V_\Theta.$$

The final inequality follows from local measure monotonicity, followed by the logarithm and the positive weighted sum. A different volume functional on a non-product region requires a separate monotonicity theorem. This is a sufficient numerical interface. Constructing the genuine vectors, establishing closure and normalization, controlling the determinant sum, and deriving the uniform IUT IV height comparison [12] remain separate obligations. Archimedean factors and infinite-packet convergence are not covered by the finite local theorem.

5. EFFECTIVE CONCENTRATION IN A SPECIFIED PELL RESIDUAL

We now restrict to positive integers b, A, B, u, v, r_0, s, X and a prime $p \geq 37$ satisfying

$$(16) \quad b = Au^2, \quad b+1 = Bv^2, \quad b+2 = 3r_0^2, \quad b+3 = s^2,$$

$$(17) \quad D = 3AB, \quad Z = b^2 + 3b + 1 = T_p(X), \quad D+1 \leq X^2, \quad X^p \leq Z.$$

Here T_p is the Chebyshev polynomial of the first kind. These equations define the specified residual packet; no reduction of general *abc* counterexamples to it is asserted. Set

$$H = \log(b+2), \quad L = \log(D+1), \quad C_0 = 12^{331776}, \quad K_0 = \log(2C_0).$$

The packet immediately yields

$$(18) \quad (D+1)^p \leq X^{2p} \leq Z^2 < (b+2)^4, \quad pL < 4H.$$

Theorem 2.2 of Bérczes–Evertse–Győry [1], specialized to $K = \mathbb{Q}$, $S = \{\infty\}$, and exponent two, gives the following published input. If $f \in \mathbb{Z}[t]$ is squarefree of degree $n \geq 3$, with coefficient height $\hat{h} = \log \max(1, |a_0|, \dots, |a_n|)$, and $f(x) = y^2$ in integers, then

$$(19) \quad \max\{h(x), h(y)\} \leq (4n)^{4096n^4} \exp(50n^4 \hat{h}), \quad h(t) = \log \max(1, |t|).$$

Proposition 5.1 (Two effective constraints). *Every packet (16)–(17) satisfies*

$$(20) \quad H \leq \exp(4300p^5), \quad H \leq 2C_0 D^{4050}.$$

Proof. First put $f_p(t) = 4T_p(t) + 5$. The relation in (17) gives $f_p(X) = (2b+3)^2$. The identities $T'_p = pU_{p-1}$ and $T_p^2 - 1 = (t^2 - 1)U_{p-1}^2$ show that a repeated root of f_p would have $T_p = \pm 1$ and $T_p = -5/4$, a contradiction. If M_n is the sum of the absolute values of the coefficients of T_n , then $M_{n+2} \leq 2M_{n+1} + M_n$ and $M_0 = M_1 = 1$. Induction gives $M_n \leq 3^n$. Thus every coefficient of f_p has absolute value at most $4 \cdot 3^p + 5 \leq 3^{p+2}$, and $\hat{h} \leq (p+2) \log 3 \leq 4p$. Since $\log(4p) \leq p$ for $p \geq 3$, (19) implies

$$H \leq \log(2b+3) \leq \exp\left(4096p^4 \log(4p) + 50p^4 \hat{h}\right) \leq \exp(4300p^5).$$

For the second estimate, set $x = b+1$ and $Y = Duvr_0$. The first three equations in (16) give

$$(21) \quad Y^2 = D(x^3 - x).$$

The cubic $D(t^3 - t)$ is squarefree and has coefficient height $\log D$. Equation (19), with $n = 3$, gives

$$\log(b+1) \leq 12^{4096 \cdot 3^4} \exp(50 \cdot 3^4 \log D) = C_0 D^{4050}.$$

Finally $b+2 \leq (b+1)^2$ for $b \geq 1$, proving the second bound. $\square$

Proposition 5.2 (Descent to the squarefree parameter). *Write $D = d\tau^2$, where d is a positive squarefree integer and $\tau \geq 1$ is an integer. Then every packet satisfies*

$$H \leq 2C_0 d^{4050}.$$

Proof. Equation (21) implies $\tau^2 \mid Y^2$. Comparing prime valuations gives $\tau \mid Y$. Write $Y = \tau y$ and cancel the nonzero factor τ^2 to obtain

$$y^2 = d(x^3 - x), \quad x = b+1.$$

The polynomial $d(t^3 - t)$ has distinct roots and coefficient height $\log d$. Equation (19) gives $\log(b+1) \leq C_0 d^{4050}$, and again $H \leq 2 \log(b+1)$. $\square$

Combining (18) and (20) gives

$$(22) \quad p \geq \left(\frac{\log H}{4300}\right)^{1/5}, \quad L < 4 \cdot 4300^{1/5} \frac{H}{(\log H)^{1/5}},$$

$$(23) \quad D \geq \left(\frac{H}{2C_0}\right)^{1/4050}, \quad p(\log H - K_0) < 16200H.$$

For example, the first upper bound follows by multiplying the lower bound for p by $L > 0$ and applying $pL < 4H$. For the final inequality use $\log H \leq K_0 + 4050 \log D$ and $\log D < L$. In a form avoiding fractional powers, the former calculation is

$$(\log H)L^5 \leq 4300(4H)^5.$$

Corollary 5.3. *Any sequence of these packets with strictly increasing b_n , with d_n the positive squarefree representative of D_n , satisfies*

$$p_n \longrightarrow \infty, \quad d_n \longrightarrow \infty, \quad D_n \longrightarrow \infty, \quad \frac{\log(D_n + 1)}{\log(b_n + 2)} \longrightarrow 0.$$

Proof. Here $H_n \rightarrow \infty$ and $\log H_n \rightarrow \infty$. Apply (22) and (23). The bound in Proposition 5.2 forces $d_n \rightarrow \infty$. $\square$

These restrictions are compatible, so they do not exclude the residual sequence. Distinct positive squarefree integers $d > 1$ give distinct quadratic fields $\mathbb{Q}(\sqrt{d})$, so these fields cannot remain in a fixed finite collection. Also $\log d_n/H_n \rightarrow 0$ because $1 \leq d_n \leq D_n$. Thus freezing either the index or the quadratic field cannot be justified for an unbounded sequence. The full argument uses (19); Lean checks only the explicit arithmetic and scalar consequences with that input stated as a hypothesis.

6. APPLICABILITY OF RECENT ELLIPTIC-CURVE ESTIMATES

6.1. A denominator lower bound. For the Frey curve $E : y^2 = x(x - a)(x + b)$ attached to an abc triple, let $Q = a^2 + ab + b^2$. Its invariant is

$$j_E = \frac{256Q^3}{(abc)^2}.$$

Let n, d be the positive numerator and denominator in lowest terms.

Proposition 6.1. *For every such triple,*

$$(24) \quad c^4 \leq 1024d, \quad n \leq 256c^6.$$

Proof. Primitivity implies $\gcd(Q, abc) = 1$. Indeed, reduction modulo a prime dividing any one of a, b, c leaves in Q the square of a nonzero other coordinate. Therefore $g = \gcd(256Q^3, (abc)^2)$ divides 256, and $(abc)^2 = gd \leq 256d$. Since $(a - 1)(b - 1) \geq 0$, we have $ab \geq c - 1$; as $c \geq 2$, this gives $2ab \geq c$. Squaring $2abc \geq c^2$ proves $c^4 \leq 4(abc)^2 \leq 1024d$. Finally $Q = c^2 - ab \leq c^2$, so $n \leq 256Q^3 \leq 256c^6$. $\square$

Pasten's 2026 small-denominator theorem [14, Theorem 1.2] applies under a bound $\text{den}(j_E) \leq A_0(\log \text{num}(j_E))^{B_0}$ with fixed positive A_0, B_0 . For Frey curves, (24) would imply

$$c^4 \leq 1024A_0(\log 256 + 6 \log c)^{B_0}.$$

A positive power of c dominates every fixed power of its logarithm. Thus only bounded c , and consequently finitely many positive triples, can satisfy this hypothesis for fixed A_0, B_0 . This is a precise restriction on direct applicability, not a counterexample to the theorem or to other Szpiro strategies.

6.2. The August 2026 Mordell estimate. The preprint [13, Theorem 1.3], submitted August 24, 2026, gives an effective bound for $y^2 = x^3 + k$, $k \neq 0$, using

$$\underline{k} = \prod_{q|k} q^{\min(2, v_q(k))}.$$

The stated estimate is

$$(25) \quad \log \max(|x|, |y|) \ll \underline{k}^{1/2} \log^2(2\underline{k}) \log(2|k|) \log(\underline{k} \log(3|k|)).$$

We record this new result without treating its publication date as a substitute for checking its hypotheses or proof.

Directly substituting $c_6^2 = c_4^3 - 1728\Delta$ and $\Delta = 16(abc)^2$ gives $k = -27648(abc)^2$. Then $k^{1/2} \ll r$, while both $\log(2|k|)$ and $\log \max(|c_4|, |c_6|)$ are comparable with h . The resulting right-hand side has the shape

$$r \log^2(2r) h \log(2r \log(3c)).$$

The height remains a multiplicative factor on the right; cancellation does not yield an upper bound for c in terms of r . The integral- j result in Theorem 1.6 of the same paper concerns the height of the *curve*, and cannot be substituted for a point-height bound.

There is also a strict obstruction to one possible transfer from (21). With $U = Dx$ and $V = DY$, it becomes $E_D : V^2 = U^3 - D^2U$, of j -invariant 1728.

Proposition 6.2. *For $D, k \neq 0$, there is no nonconstant algebraic morphism over $\overline{\mathbb{Q}}$ from E_D to the nonsingular Mordell curve $M_k : v^2 = u^3 + k$.*

Proof. Their geometric endomorphism algebras are respectively $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$: the automorphisms $(x, y) \mapsto (-x, iy)$ and $(u, v) \mapsto (\zeta_3 u, v)$ give these quadratic actions, and in characteristic zero the endomorphism algebra of an elliptic curve has dimension at most two over $\mathbb{Q}$. A nonconstant morphism, translated to send the origin to the origin, is an isogeny. An isogeny induces an isomorphism of rational endomorphism algebras by conjugation. The two imaginary quadratic fields are not isomorphic, a contradiction; see also [15, Chapters III and VI]. $\square$

The proposition excludes this morphism strategy, even if k depends on D . It does not exclude other Diophantine encodings or extending the new logarithmic-form methods to a different family.

7. FORMAL VERIFICATION AND REMAINING OBLIGATIONS

The companion development uses Lean 4.32.0 and the repository’s pinned dependencies. The new files are:

Module (within IUTThreeClosures)	Checked mathematical scope
SmoothLowOmegaCounting	Actual prime-power encoding, finite cardinality bounds, and the finite first moment.
SignedPrimeSupport	Actual prime-exponent sums and the coupled equivalence with the original target.
SignedEndpointDyadicObstruction	The genuine dyadic triples and failure of a separate-endpoint slope below one.
ReachableTensorLatticeCriterion	Unit-determinant span certificate, symmetric span and its missing off-diagonal tensor.
FreyPelleEffectiveResidual Corridor20260830	The integral elliptic identity, integer square-factor descent, explicit scalar implications, and the Frey denominator bound.

These modules introduce no new mathematical axiom. The validation record lists their exact commands, results, and kernel dependency reports. The pre-existing signed-layer module required an explicit import, closure of an elementary algebra branch, and the correct order of unfolding before rewriting; these were repaired without changing its statements.

The formal scope is deliberately narrower than the full paper. Theorem 2.3 and Corollary 2.4 use Younis and classical smooth-number asymptotics in their mathematical proofs; those external analytic theorems are not formalized here. The complete local Haar formula in Theorem 4.1, the general independent-action theorem, the effective bound (19), the asymptotic consequences of the Pell bounds, and the endomorphism-algebra argument also retain their stated paper-proof

boundary. A Lean implication with an explicit height hypothesis is not a Lean proof of that hypothesis.

The surviving proof target is a uniform estimate for the coupled expression (15), or a source-derived theorem strong enough to imply it. The IUT route still needs genuine reachable output data, compatible normalizations and the global comparison. The analytic disproof route still needs a height-unbounded family satisfying one fixed radical exponent gap; the finite-support bound neither constructs nor forbids such a sparse family. The Pell residual inequalities currently admit a nonempty numerical corridor and do not apply to all abc triples. No unconditional closed term of type `ABCConjecture`, or of its negation, is produced by this work.

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